Student number_____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

The First Semester of 2022—2023 Year

Applied Calculus I Final Examination

(Closed book Time 120 minutes)

Question number	I	II	III	IV	V	Total score
Score						
Grader						

I. Completions (5 items, 3 points each)

Score	

1. If $\lim\limits_{x\to\infty}\dfrac{ax^2-x+\sin x}{2x^2-\cos x}=2$, then $a=$_____ .

2. Assume that $f(x)$ is an even function and $f'(0)$ exists. Then $f'(0)=$_____.

3. The equation of the tangent line to the parabola $y=x^2-8x+9$ at $(1,2)$ is _____.

4. The number of positive roots of the equation $x^5+2x-2=0$ is_____.

5. The absolute maximum of $y=xe^{-x^2/8}$ over the interval $[-1,4]$ is _____ .

II. Choices (5 items, 3 points each)

Score	

6. If $f(x)$ has second derivative and $y=f\left(f(x)\right)$, then $y''=$ ( )

A. $f'\left(f(x)\right)\left(f'(x)\right)^2+f''\left(f(x)\right)f'(x)$. B. $f''\left(f(x)\right)\left(f'(x)\right)^2+f'\left(f(x)\right)f''(x)$.

C. $f'\left(f(x)\right)\left(f'(x)\right)^2+f'\left(f(x)\right)f''(x)$. D. $f''\left(f(x)\right)\left(f'(x)\right)^2+f'\left(f(x)\right)f'(x)$.

7. Assume f has a continuous second derivative over an interval containing a, then $\lim_{h\to 0}\frac{f(a+2h)+f(a-2h)-2f(a)}{2h^2}=$ ( )

A. $2f'(a)$

B. $4f''(a)$.

C. $4f'(a)$.

D. $2f''(a)$.

8. If the curve $y=3x^5-2x^3$, then ( )

A. y has three point of inflection.

B. y has three critical points.

C. y has two points of inflection.

D. y has one critical points.

9. If $f''(x)>0, x\in[0,1]$, then ( )

A. $f'(1)>f'(0)>f(1)-f(0)$.

B. $f'(1)>f(1)-f(0)>f'(0)$.

C. $f(1)-f(0)>f'(1)>f'(0)$.

D. $f'(1)>f(0)-f(1)>f'(0)$.

10. If $h(x)=\sqrt{4+3f(x)}$, where $f(1)=7$ and $f'(1)=4$, then $h'(1)=$ ( )

A. $\frac{1}{3}$.

B. $\frac{2}{3}$.

C. $\frac{3}{4}$.

D. $\frac{6}{5}$.

Score	

III. Calculations (6 items, 8 points each)

11. Find the derivative of $y=\sqrt{x+\sqrt{x+\sqrt{x}}}$.

12. Let $y=\sin^4 x+\cos^4 x$, Calculate $y^{(n)}$.

(Hint: $(\sin x)^{(n)}=\sin(x+\frac{n\pi}{2})$ and $(\cos x)^{(n)}=\cos(x+\frac{n\pi}{2})$).

13. Calculating $\lim\limits_{x\to 0}\left(\frac{1}{x^2}-\frac{1}{x+\tan x}\right)$. (Hint: Apply the Taylor formula.)

14. Calculating $\lim\limits_{x\to 0^+}\cot x^{\frac{1}{\ln x}}$.

15. If $x^2 + xy + y^3 = 1$, find the value of y''' at the point where $x = 1$.

16. Find the Maclaurin formula (with the Lagrange remainder) for $\cos^2 x$.

Score	

Ⅳ. Proofs（2 items, 7points each）

17. Assume that $f(x)$ is continuous over $[0,1]$ and $f'(x)$ exists for x in $(0,1)$. Moreover, $f(0)=f(1)=0, f(\frac{1}{3})=1$. Prove that there is an $\eta \in (0,1)$ such that $f'(\eta)=1$.

18. Show that $\sqrt{1+x}<1+\frac{x}{2}$ if $x>0$.

V. Completion（8 points）

Score	

Definition 1: Assume $f(x)$ *is well-defined over an interval* (a,b). *If the equality*

$$f\left(\frac{x_1+x_2}{2}\right)<\frac{f(x_1)+f(x_2)}{2}$$

holds for any $x_1,x_2\in(a,b)$, *then we say* $f(x)$ *is concave up.*

19. Let $f(x)=\dfrac{(x-1)^3}{(x+1)^2}$.

(1) Find the intervals in which $f(x)$ is strictly increasing.

(2) Find the intervals in which $f(x)$ is concave up.

(3) Find the formula for the oblique asymptote of $f(x)$.